2. Literature Review & Competitive Analysis

UNSTUCK (AI Automation / Freelance Intelligence Platform)

TRACK & PROGRAM	TARGET MILESTONE	DOCUMENT CODE	ACADEMIC CYCLE
DEPI Technical Track	16 October 2026	DEPI-DOC-02 (M2)	October – December 2026

1. Market Landscape & Problem Formulation

The freelance gig economy has grown rapidly across the Middle East and globally. Platforms such as **Mostaql, Khamsat,** Nafezly and Freelancer publish thousands of diverse project listings daily. However, the ecosystem remains intensely fragmented:

- **Siloed Information:** Freelancers are forced to manually maintain separate browser tabs across 4 to 5 platforms, constantly refreshing feeds to capture new opportunities.
- **Time Inefficiency:** An average freelancer spends 90 to 180 minutes daily performing routine search and screening tasks rather than billable engineering work.
- **Asymmetric Skill Matching:** Traditional search filters rely on naive keyword matching, which frequently overlooks transferable competencies or promotes irrelevant postings.
- **Proposal Fatigue:** Crafting tailored, high-converting bids under tight time limits leads to freelancer burnout and lower win rates.

2. Comparative Analysis of Existing Solutions

System Type	Representative Solutions	Technical Limitations	Our Project Advantage
Native Platform Feeds	Mostaql, Khamsat internal search.	Siloed to one platform; opaque recommendation algorithms; no cross-platform ranking.	**Unified Repository:** Consolidates multi-platform listings into a normalized datastore.
Generic Job Boards	LinkedIn, Indeed, Bayt.	Optimized for long-term full-time roles; ignores gig budgets, project durations, and bidding models.	**Gig-Specific Schemas:** Tracks milestone pricing, bid counts, and freelance skill tags.
Basic Web Scrapers / RSS Aggregators	Custom Python scripts, basic feed readers.	Fragile; broken by anti-bot changes; high duplication; no AI semantic reasoning.	**Anti-Bot & AI Pipeline:** Headless browser bypass, hash deduplication, and LLM skill matching.
Commercial Auto-Apply Bots	Generic third-party browser extensions.	Blind spamming without skill verification; triggers account bans; poor bid quality.	**Intelligent Auto-Bidding:** Explainable skill-gap scoring, custom copywriting, and human-in-the-loop oversight.

3. Technological Foundations & Architectural Decisions

> **Core Engineering Principle: Orchestrated, Resilient Automation**
> The project combines workflow orchestration (n8n), a scraping technology still under consideration, SQL data modeling, and Large Language Models (LLMs) into an integrated multi-agent architecture.

Architecture Layer	Selected Technology	Justification & Comparative Advantage
Orchestration Engine	n8n (Workflow Automation)	Decouples scheduling, retry logic, and webhook handling from backend servers; visual debugging; robust error logging.
Data Harvesting	Under consideration	Technology selection remains open; the approach will be evaluated for with site compatibility and reliability assessed during selection.
Datastore & Caching	SQL Database	Relational data storage, indexing for filtering, and unique hash constraints for 0% duplication, and Redis caching for sub-second queries.
Backend REST API	Python FastAPI	High-performance asynchronous architecture, native OpenAPI documentation, and seamless integration with AI/Pydantic schemas.

Architecture Layer	Selected Technology	Justification & Comparative Advantage
Web Frontend UI	React + Vite	Component-based modular architecture, dynamic RTL/LTR support, responsive layouts, and Recharts for market analytics visualizations.
AI Matching & Copywriting	LLM APIs (OpenAI / Gemini)	Enables deep semantic skill reasoning beyond keywords, explains skill gaps transparently, and drafts personalized client proposals.

4. Mentor Feedback & Design Evolution

During project evaluation sessions, academic mentors and technical instructors provided strategic guidance that shaped our architectural evolution:

- **Rigor Over Premature Complexity:** Mentors emphasized building an unbreakable data ingestion and deduplication foundation before introducing multi-agent bidding.
- **Explainability over Black-Box AI:** The evaluation committee required that AI matching not merely output an arbitrary percentage score, but clearly itemize which skills matched, which are missing, and provide rationales.
- **Human-in-the-Loop Oversight:** While full autonomous bidding is built into the architecture, a flexible manual approval toggle was implemented to guarantee freelancer confidence and platform compliance.

5. Suggested Improvements Incorporated into Design

Area	Design Improvement Implemented	Technical & Operational Impact
Decoupled Extraction	Modular scraper nodes within n8n workflows.	If a source website modifies its DOM structure, only that specific adapter is updated without affecting the database or UI.
Hash Deduplication	MD5/SHA256 content hashing on (Platform + URL + Title).	Guarantees 0% duplicate records across continuous harvesting cycles regardless of ingestion frequency.
Token Cost Mitigation	Rule-based pre-filtering and Redis match caching.	Prevents unnecessary LLM API calls on irrelevant jobs, reducing token expenditure by over 70%.
Market Analytics	Aggregated database queries visualizing trends and budgets.	Empowers freelancers to identify in-demand skills and lucrative project categories in regional markets.

6. Alignment with DEPI Graduation Criteria

DEPI Evaluation Category	Weight	Compliance in Our Project Implementation
1. Documentation & Analysis	25%	Comprehensive specifications across all 6 milestones, RACI matrix, Gantt roadmap, and literature benchmarking.
2. System Analysis & Design	25%	Modular multi-tier architecture: Ingestion (n8n / scraping technology under consideration) -> SQL Database -> Backend (FastAPI) -> UI (React + Vite).
3. Code Quality & Implementation	35%	Clean asynchronous Python codebase, Pydantic type validation, Git version control, and Docker containerization.
4. Testing, QA & Final Presentation	15%	Automated test suites, end-to-end verification scripts, executive presentation deck, and live demonstration.